EXPA
2/8/26

MLP Classification on the Iris Datasets using Different Hidden Layers and Activation Function

AIM:

To develop an MLP model for Iris classification by experimenting with different hidden layers and activation function and evalute its performance using accuracy and loss.

ALGORITHM:

* Import required python libaries
* Load the Iris dataset and separate features and targets labels.
* Encode the target labels into categorical fo
* Split the data into training and testing sets.
* Standardize the input features using StandardScale
* Create an MLP with different hidden layers and categorical cross activation functions.
* compile the model using adam optimizer and Categorical cross - entropy loss.
* Train the model using the traning data.
* Evaluate each model using accuracy and loss on the test data.
* Compare the results of different MLP models and identify the best performing model.

CODE:

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical
iris = load_iris()
x = iris.data
y = iris.target
print("Features Shape:", x.shape)
print("Target Shape:", y.shape)
y = to_categorical(y)
x_train, x_test, y_train, y_test = train_test_split(
    x, y,
    test_size = 0.2,
    random_state = 42
)
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
model1 = Sequential()
model1.add(Dense(8, activation='relu', input_shape=(4,)))
model1.add(Dense(3, activation='softmax'))
model1.compile(
    optimizer='adam',
```

OUTPUT:

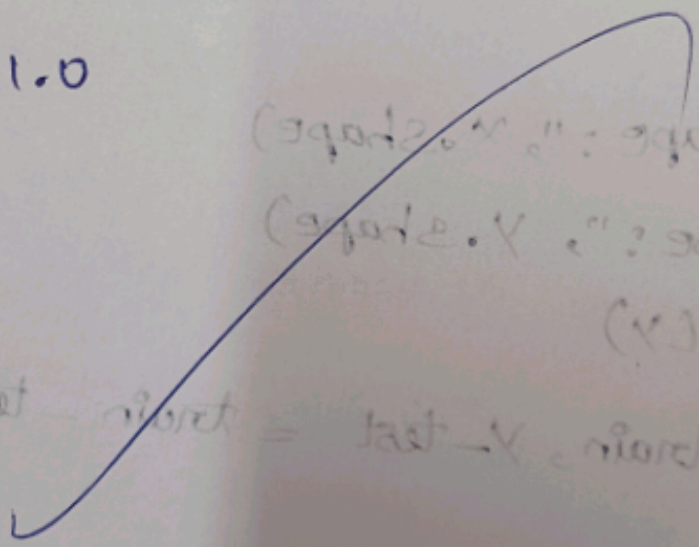
Features Shape : (150, 4)

Target Shape : (150,)

Experiment 1

Loss : 0.088149152

Accuracy : 1.0



```
history1 = model1.fit(
    x_train,
    y_train,
    epochs=100,
    batch_size=8,
    verbose=0
)
loss1, acc1 = model1.evaluate(x_test, y_test, verbose=
print("Experiment 1")
print("Loss:", loss1)
print("Accuracy:", acc1)
```

RESULT:

Thus, The MLP model successfully classified the Iris dataset. Different hidden layers and activat functions were tested, and their performance was evaluated using accuracy and Loss.